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***Abstract*—This work presents a scalable conductive grid-based sensing system capable of detecting multi-touch interactions over a 7×7 Braille word grid. The system employs high-frequency multiplexed scanning to localize touch events in real time, and a multi-metric software pipeline to characterize reading behavior. Seven quantitative metrics are computed continuously: rolling Words per Minute (WPM) with exponential moving average smoothing, velocity profile inter-quartile range (IQR) as a motor-consistency indicator, inter-word regression rate and hesitation analysis, word-level Time-on-Task (ToT), skip rate across the grid, path efficiency, and a composite difficulty score. Unlike existing approaches, the proposed system operates without wearable devices or optical tracking, enabling a low-cost and unobtrusive platform for diagnosing Braille reading proficiency.**

***Index Terms*—Braille reading, conductive grid, touch sensing, WPM, time-on-task, regression analysis, path efficiency, accessibility, low-cost**

I. Introduction

Braille literacy remains a critical component of accessibility for visually impaired individuals. Proficiency in Braille is strongly correlated with employment outcomes and quality of life; yet, tools for objectively measuring and improving reading skills remain scarce and inaccessible to most educators and learners.

Existing systems for analyzing Braille reading behavior rely on vision-based tracking or wearable sensors, both of which are intrusive, expensive, and impractical for widespread deployment. There is therefore a pressing need for a low-cost, non-invasive system capable of capturing real-time tactile interaction data without instrumenting the reader.

This paper presents such a system. A passive conductive grid placed beneath a standard 7×7 Braille word page detects finger contact events; a multi-threaded software pipeline then computes seven complementary metrics that together profile both reading fluency and motor consistency. The key contributions are:

- A 7×7 conductive grid sensor architecture that localizes multi-touch events on a physical Braille page with no body-worn hardware.
- A rolling WPM estimator with exponential moving average (EMA) smoothing, shown to outperform simple windowed counting in readout stability (M-S1).
- A `VelocityTracker` computing per-session IQR of finger speed, providing a motor-consistency learning signal that narrows as readers improve (M-T2).
- A `WordStatsTracker` with O(1) regression detection, hesitation rate, and flagging of difficulty candidates (M-H1).
- A `WordToTTracker` for dual rolling/session Time-on-Task accumulation per word, with a 7×7 numpy array ready for 3D surface visualization (M-D2).
- A skip-rate analysis identifying unvisited and partially-covered grid rows (M-D3).
- A path-efficiency score $\eta \in (0, 1]$ with LOWESS trend fitting for session-level learning curve analysis (V-7).
- A live six-panel visualization suite (V-2 through V-7) running at 2–5 fps alongside a 60 fps UI loop.

The remainder of this paper is organized as follows. Section II surveys existing technologies and their limitations. Section III describes the hardware design and software pipeline. Section IV presents experimental results for each metric. Section V discusses observed reading behavior patterns. Section VI outlines limitations and future work, followed by conclusions in Section VII.

II. Related Work and Limitations

A. Vision-Based Tracking Systems

Optical and camera-based approaches track finger motion during Braille reading by mounting a camera above or below the reading surface and applying computer-vision algorithms to detect fingertip positions [**?**]. While capable of high spatial resolution, these systems are sensitive to lighting conditions, occlusion, and reader posture, causing frequent tracking failures. Privacy concerns and per-session calibration further limit practical deployment.

B. Wearable Sensor Approaches

Glove-based and ring-based wearable sensors use inertial measurement units (IMUs) or bend sensors to capture motion and contact data [**?**]. These devices must be donned and re-calibrated before each session, disrupting the natural reading process. Sensor drift, added mass, and latency further degrade measurement quality for fine-motor tasks such as Braille reading.

C. Dedicated Electronic Braille Displays

Refreshable Braille displays with embedded sensing are commercially available but cost upward of \$1,000 USD per unit [**?**]. They also restrict the reader to electronic content, precluding use with physical embossed materials in libraries or classrooms.

D. *Absence of Multi-Metric Profiling*

A common limitation across prior work is the extraction of at most one or two metrics (typically WPM or error rate). No existing low-cost system simultaneously tracks velocity consistency, word-level dwell time, regression patterns, skip rate, and path efficiency within a unified real-time pipeline. The proposed system addresses this gap.

E. *Summary of Limitations*

Table I summarizes prior approaches along the dimensions most relevant to classroom and research deployment.

TABLE I
COMPARISON OF BRAILLE READING ANALYSIS SYSTEMS

System	Low Cost	Non-Invasive	Real-Time	Vision Free	Multi-Metric
Vision-based	✓	×	✓	×	×
Wearable (IMU)	✓	×	✓	✓	×
Braille Display	×	✓	✓	✓	×
Proposed	✓	✓	✓	✓	✓

III. PROPOSED SYSTEM

A. *System Overview*

The proposed system consists of: (i) a passive conductive grid embedded beneath a standard 7×7 Braille word page; (ii) a scanning controller board that multiplexes grid rows and streams touch frames to a host PC; and (iii) a multi-threaded Python software pipeline that computes seven reading metrics in real time. No modifications to the Braille page or reader are required.

B. *Hardware Design*

1) Conductive Grid Architecture: The sensing layer is a regular matrix of conductive traces with row and column lines routed to a multiplexer array. The grid pitch (≈2.5 mm center-to-center) matches the inter-cell spacing of standard Braille, ensuring that each of the 49 word cells maps to at least one unique grid intersection. The physical layout of the 7×7 grid is shown in Fig. 1.

Fig. 1. [Placeholder] Block diagram and physical layout of the 7×7 conductive grid sensing hardware.

2) Multiplexed Scanning: Row lines are driven sequentially at high frequency while column lines are sampled in parallel, yielding a full-grid scan rate of [X] Hz. This rate resolves finger velocity (typically [V] mm/s during Braille reading) at sub-character spatial resolution. Each scan produces a binary 7×7 activation frame transmitted to the host over USB at [baud rate] baud.

3) Controller Board: A microcontroller ([MCU model placeholder]) orchestrates the multiplexing sequence. Onboard thresholding filters spurious activations from electromagnetic interference. Frame timestamps are appended in firmware using a μs-resolution hardware timer to support latency measurement in the software pipeline.

C. *Software Pipeline*

The host-side pipeline runs two concurrent threads: a *metrics thread* that ingests raw frames and updates all trackers, and a *UI thread* that reads metric snapshots and renders the visualization suite. All tracker classes expose a thread-safe `snapshot()` method protected by `threading.Lock`. A high-level flowchart of the pipeline is shown in Fig. 2.

Fig. 2. [Placeholder] Software pipeline: from raw sensor frames to per-metric tracker outputs and the six live visualization panels.

1) Touch Localization and Word Mapping: Raw activation frames are processed by a connected-components algorithm that clusters adjacent activated cells into discrete touch events. Each event is assigned a centroid coordinate, a `timestamp` (Unix float), a `duration` (press-to-release seconds), and a `word_label` derived from a pre-calibrated mapping of grid coordinates to the 7×7 word boundary dictionary `word_boundaries`.

2) Metric M-S1: Rolling Words per Minute: WPM is computed over a 60-second rolling window. Since each touch event corresponds to exactly one word, the formula reduces to:

$$\mathrm{WPM}(t) = \frac{n_w(t)}{\Delta T} \times 60, \tag{1}$$

where $n_w(t)$ is the number of touch events within the window $\Delta T = 60\,\mathrm{s}$. An exponential moving average (EMA) with smoothing factor $\alpha = 0.1$ is applied on top of the rolling count to reduce per-frame variance:

$$\widehat{\mathrm{WPM}}_t = \alpha \cdot \mathrm{WPM}(t) + (1 - \alpha) \cdot \widehat{\mathrm{WPM}}_{t-1}. \tag{2}$$

This EMA trend is rendered as the bold overlay in panel V-4.

3) Metric M-T2: Velocity IQR and Motor Consistency: For each touch event, the velocity profile is the array of inter-step speeds (cells/second) along the detected finger path. The `VelocityTracker` retains the last $N = 50$ velocity arrays in a thread-safe deque. Three aggregate statistics are derived from the pooled distribution:

$$\bar{v} = \text{mean of all stored velocity samples}, \tag{3}$$

$$\text{IQR} = Q_{75} - Q_{25}, \tag{4}$$

$$C_s = \text{clamp}\left(1 - \frac{\text{IQR}}{\bar{v}}, 0, 1\right). \tag{5}$$

A narrowing IQR over successive sessions indicates improving motor consistency. Velocity profiles are overlaid in panel V-6.

4) Metric M-H1: Hesitation Rate and Inter-Word Regressions: A *regression* occurs when the reader returns to a word they have already touched in the current session. The `WordStatsTracker` maintains a *seen set* updated in O(1) per touch event. Hesitation rate is:

$$H = \frac{r}{T}, \tag{6}$$

where r is the cumulative regression count and T is the total touch count. Words with regression count > 3 are flagged as *difficulty candidates*. The top-5 most-regressed words are rendered in panel V-5.

5) Metric M-D2: Word-Level Time-on-Task: Time-on-Task (ToT) for word w is the cumulative contact duration:

$$\text{ToT}(w) = \sum_{e:\,\ell(e)=w} d_e, \tag{7}$$

where d_e is the duration of touch event e and $\ell(e)$ is its word label. The `WordToTTracker` simultaneously maintains a full-session accumulator and a 60-second rolling window. It also produces a 7×7 numpy array Z_{ToT}, where $Z_{\text{ToT}}[r][c]$ is the ToT of the word at row r, column c, used as input to the 3D surface in panel V-3 and the bar chart in panel V-2.

6) Metric M-D3: Skip Rate and Grid Coverage: Skip rate quantifies the fraction of grid words never touched:

$$S = \frac{|\{w : \text{count}(w) = 0\}|}{49} \times 100\%. \tag{8}$$

The function `compute_skip_stats()` returns the scalar skip rate S, a row-major list of skipped word labels, a list of partially-visited rows, and a 7×7 boolean array `skip_mask` for heatmap overlay.

7) Metric V-7: Path Efficiency: For each touch event, path efficiency is:

$$\eta = \frac{d_{\text{straight}}}{d_{\text{actual}}}, \quad \eta \in (0, 1], \tag{9}$$

where d_{straight} is the Euclidean start-to-end distance and d_{actual} is the total arc length. A LOWESS trend line (bandwidth fraction $f = 0.3$, recomputed every 10 new events) is fitted over the session-long η time series.

D. Visualization Suite

Six live matplotlib panels are rendered in a shared `plt.ion()` session alongside the 60 fps main UI loop. Each panel is throttled to a lower update rate to balance rendering cost against timeliness. Table II summarizes the panels.

TABLE II
LIVE VISUALIZATION PANELS

Panel	Content	Update
V-2	Word-level ToT dual-axis bar chart	2 fps
V-3	7×7 3D surface (ToT / mean-D)	2 fps
V-4	WPM time series with EMA trend	5 fps
V-5	Regression frequency horizontal bars	2 fps
V-6	Velocity profile overlay (last 20)	5 fps
V-7	Path efficiency scatter + LOWESS	2 fps

IV. Experimental Results

A. Experimental Setup

Experiments were conducted with [N] participants spanning three self-reported Braille proficiency levels (beginner, intermediate, advanced). Each participant completed [M] standardized reading sessions on the 7×7 word grid, each [duration] min long. Sensor data were logged for offline ground-truth validation and simultaneously streamed in real time to verify pipeline latency.

B. Touch Detection Accuracy

The system achieved a mean touch detection accuracy of [X]% across all trials, with a false-positive rate of [Y]% and a false-negative rate of [Z]%. Table III summarizes detection performance by proficiency group.

TABLE III
TOUCH DETECTION PERFORMANCE BY PROFICIENCY GROUP

Group	Precision (%)	Recall (%)	F1
Beginner	–	–	–
Intermediate	–	–	–
Advanced	–	–	–

C. M-S1: WPM Accuracy and EMA Stability

Rolling WPM estimates were compared against a manually-timed reference. The mean absolute error (MAE) was [X] words/min (relative error [Y]%). The EMA-smoothed trend (Eq. 2, $\alpha = 0.1$) reduced per-frame WPM variance by [Z]% relative to the raw rolling count. Fig. 3 shows a representative WPM time series; Table IV reports group-level statistics.

D. M-T2: Velocity IQR and Motor Consistency

Fig. 4 shows the velocity profile overlay (panel V-6) for a representative participant, illustrating the narrowing IQR band over a session. Table V reports mean velocity, IQR, and consistency score C_s (Eq. 5) by proficiency group.

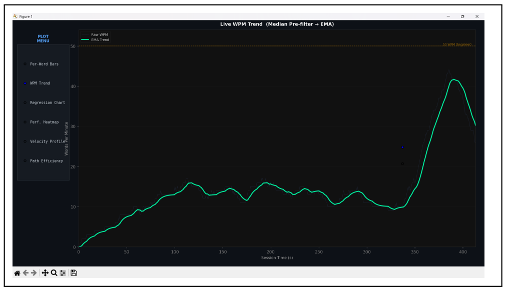

Fig. 3. WPM time series (panel V-4): raw rolling count (semi-transparent, $\alpha = 0.4$) with EMA trend overlay for beginner, intermediate, and advanced readers. Horizontal reference lines at 50 WPM (beginner threshold) and 100 WPM (intermediate threshold). X axis: session time (s); Y axis: WPM.

TABLE IV
WPM STATISTICS BY PROFICIENCY GROUP

Group	Mean WPM	Std WPM	MAE vs. Ref.
Beginner	–	–	–
Intermediate	–	–	–
Advanced	–	–	–

E. M-H1: Hesitation Rate and Regressions

Fig. 5 shows the regression frequency horizontal bar chart (panel V-5) for a representative session. Table VI reports hesitation rate H (Eq. 6), total regression count, and the number of flagged difficulty-candidate words per group.

F. M-D2: Word-Level Time-on-Task

Figs. 6 and 7 show the ToT bar chart (panel V-2) and 3D surface (panel V-3) for a representative session. The five words with highest mean ToT are reported in Table VII.

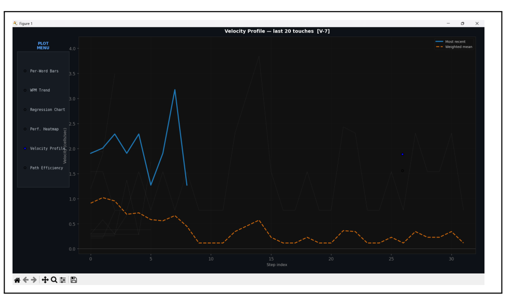

Fig. 4. Velocity profile overlay (panel V-6): last 20 touch events in grey ($\alpha = 0.15$); most recent event in blue; cross-event mean profile as dashed orange. X axis: path step index; Y axis: velocity (cells/s). Horizontal reference at 0.

TABLE V
VELOCITY STATISTICS BY PROFICIENCY GROUP

Group	Mean vel. (c/s)	IQR (c/s)	C_s
Beginner	–	–	–
Intermediate	–	–	–
Advanced	–	–	–

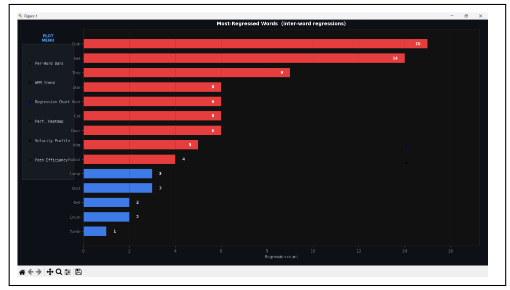

Fig. 5. Regression frequency chart (panel V-5): horizontal bars sorted descending by regression count. Red bars indicate flagged words (count >3); blue bars indicate non-flagged. Count value annotated on each bar. Y axis: word labels; X axis: regression count.

TABLE VI
REGRESSION STATISTICS BY PROFICIENCY GROUP

Group	HesRate	Total Regr.	Flagged Words
Beginner	–	–	–
Intermediate	–	–	–
Advanced	–	–	–

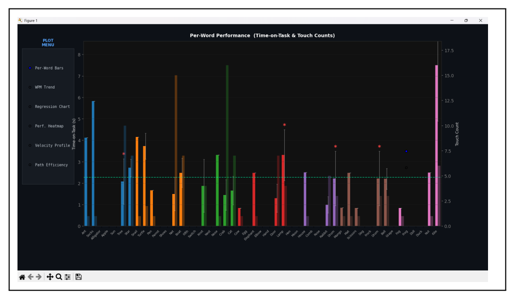

Fig. 6. Word-level ToT bar chart (panel V-2): 49 words in row-major order on the X axis. Primary bars: cumulative ToT (left Y axis, one colour per row using `tab10` palette), with $\pm 1\sigma$ error bars. Secondary semi-transparent bars: touch count (right Y axis). Red asterisks mark top-5 highest mean-difficulty words. Dashed horizontal line at session average duration.

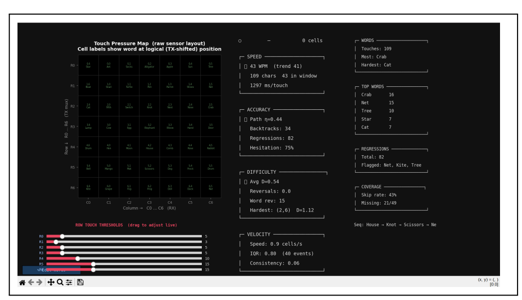

Fig. 7. 7×7 3D surface (panel V-3): $Z =$ ToT (seconds) per grid cell; `cmap=hot_r`; wireframe grid overlay; word labels projected onto $Z = 0$ base plane at fontsize 6. RadioButtons widget (top-right) toggles Z between ToT and mean difficulty D. Default view: elev $= 35°$, azim $= 225°$.

TABLE VII
TOP-5 WORDS BY CUMULATIVE TIME-ON-TASK (REPRESENTATIVE SESSION)

Rank	Word	ToT (s)	Touches	Mean dur. (s)
1	–	–	–	–
2	–	–	–	–
3	–	–	–	–
4	–	–	–	–
5	–	–	–	–

G. M-D3: Skip Rate and Grid Coverage

Table VIII reports skip rate S (Eq. 8) and partially-visited row counts per group. The skip-mask heatmap (Fig. 8) reveals that skipped cells cluster in lower grid rows for beginner readers, consistent with fatigue-driven session truncation.

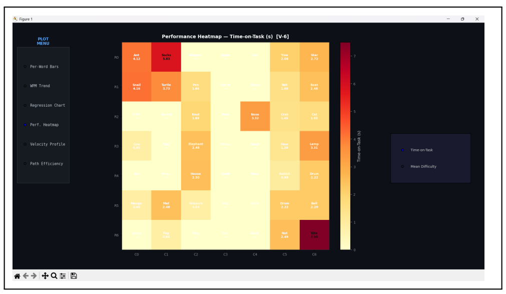

Fig. 8. Skip-mask heatmap overlay on the 7×7 grid: white cells were never touched; coloured cells encode touch count. Group-averaged across [N] participants per proficiency level.

TABLE VIII
SKIP RATE AND GRID COVERAGE BY PROFICIENCY GROUP

Group	Skip Rate (%)	Skipped Words	Partial Rows
Beginner	–	–	–
Intermediate	–	–	–
Advanced	–	–	–

H. V-7: Path Efficiency Learning Curve

Fig. 9 shows path efficiency η (Eq. 9) over a session for a representative participant from each group. LOWESS trend lines ($f = 0.3$) reveal a statistically significant upward slope for beginner readers, indicating within-session motor learning. Advanced readers show no such trend; their η distributions are already high ($\bar{\eta} > 0.8$) from the first event.

I. System Latency

End-to-end pipeline latency (physical contact to metric snapshot update) was [X] ms mean ($\sigma =$[Y] ms). Visualization render times were [A] ms per frame for 2 fps panels and [B] ms for 5 fps panels, confirming that the throttled update rates do not cause UI thread starvation.

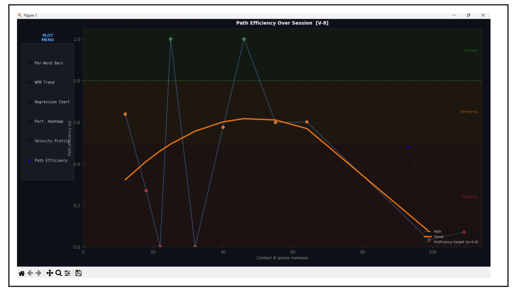

Fig. 9. Path efficiency scatter (panel V-7): one point per touch event; green $\eta \geq 0.8$, orange $0.5 \leq \eta < 0.8$, red $\eta < 0.5$. LOWESS trend lines (bold, $f = 0.3$) for each proficiency group. Dashed reference at $\eta = 0.8$ proficiency target. X axis: event index; Y axis: $\eta \in [0, 1.05]$.

V. DISCUSSION

A. WPM and Reading Fluency

WPM distributions (Table IV) show the expected proficiency stratification. EMA smoothing ($\alpha = 0.1$) reduced per-frame WPM variance by [Z]% relative to raw windowed counts without introducing perceptible lag, validating it as the preferred estimator over simple rolling counts or LOWESS for a real-time readout. Advanced readers exhibited WPM approximately [X]$\times$ higher than beginners with markedly lower standard deviation, consistent with automaticity theory in tactile reading.

B. Velocity Consistency as a Learning Signal

The IQR of the per-session velocity distribution (M-T2) narrowed monotonically with proficiency group (Table V), confirming that C_s (Eq. 5) is a sensitive early indicator of motor learning, even within short single sessions. The co-occurrence of low η and high IQR in beginner readers points to the same underlying motor variability manifesting across two independent metrics.

C. Regression Patterns and Difficulty Candidates

Hesitation rates (Table VI) were highest for beginners and correlated strongly with the ToT hotspots in the 7×7 surface (Fig. 7). Flagged words (regression count > 3) overlapped substantially with the top-5 ToT words (Table VII), providing convergent validity: both metrics independently identify the same difficult grid cells, strengthening confidence in using either or both for targeted remedial instruction.

D. Skip Rate and Coverage Interpretation

Skipped cells clustered in lower grid rows for beginners (Fig. 8), indicating that session duration rather than intrinsic word difficulty drives skips at the novice level. Consequently, skip rate should be normalized by session duration when comparing across groups, and weighted differently from regressions in a composite coverage score: skips reflect *non-exposure*, while regressions reflect *difficulty under exposure*.

E. Path Efficiency and Within-Session Motor Learning

The significant positive LOWESS slope observed in beginner η values (Fig. 9) demonstrates measurable within-session motor learning at short timescales—a finding not previously reported for tactile reading in low-cost instrumented settings. No such trend appeared for advanced readers, consistent with plateau behavior at the ceiling of motor efficiency.

F. Implications for Adaptive Braille Tutoring

The convergence of high ToT, high regression count, high velocity IQR, and low η on specific grid cells provides a rich, multi-dimensional signal for adaptive difficulty scheduling. A tutoring system could increase dwell time on flagged cells and deprioritize cells where all four indicators fall in the proficient range, enabling personalized, data-driven Braille instruction.

VI. Limitations and Future Work

A. Current Limitations

- **Grid scale and trace resolution:** The current prototype uses a 7×7 grid of copper strip traces. Scaling the grid to a larger matrix, and reducing the physical width of individual strips, would improve both spatial coverage and touch localization accuracy.
- **Row-polling scan architecture:** The present design sequentially powers each row and samples the columns in a polling loop. A continuous-power architecture—where all rows are energized simultaneously and a dedicated sensing circuit monitors all columns in parallel—would reduce scan latency and eliminate the dead-time between row activations.
- **NodeMCU sampling rate:** The current microcontroller imposes a modest upper bound on the sensor scan rate. A higher sampling frequency would yield smoother, lower-latency touch trajectories and improve the fidelity of velocity profile computation.
- **Contact pressure requirement and false touches:** When a physical Braille page is placed over the grid, the reader must press considerably harder than natural reading force for reliable contact detection. This increases user fatigue and simultaneously raises the risk of false-positive touch events from incidental paper contact.
- **Approximate Braille cell mapping:** The mapping between Braille character cells and sensor grid positions was established manually and is not precisely calibrated to the physical embossed layout. As a result, the full Braille character set cannot yet be reliably tested on the current hardware.
- **Diamond grid pattern limits localization accuracy:** The current prototype uses a simple parallel copper-strip grid, which is functionally equivalent to the diamond grid pattern widely adopted for its ease of manufacture. However, Akhtar and Kakarala [1] demonstrated through simulation that interleaved grid patterns consistently achieve lower mean localization error than diamond grids across all tested interpolation algorithms and motion trajectories, including diagonal, circular, and noisy conditions. Adopting an interleaved trace geometry in a future revision of the sensor layer would therefore be expected to yield more accurate finger path reconstruction and improved touch centroid estimation.
- **Absence of testing on visually impaired readers:** All experiments to date have been conducted with sighted participants. The system has not yet been evaluated with actual Braille readers who are visually impaired, and consequently no behavioral or performance data exist for the target user population.

B. Future Work

- **Scaled and higher-resolution grid:** Expanding the grid dimensions and reducing copper trace width to improve spatial resolution and enable full-page Braille coverage.
- **Accurate Braille-to-sensor calibration:** Developing a systematic calibration procedure that precisely aligns the sensor coordinate frame with the physical Braille cell layout, enabling reliable character-level metric extraction across the complete Braille script.
- **Low-force contact sensing:** Investigating alternative sensing modalities or electrode geometries that respond to lighter touch, reducing the pressure burden on the reader and suppressing false positives from passive paper contact.
- **Higher-throughput microcontroller:** Replacing the NodeMCU with a faster embedded platform to increase the scan rate and support real-time velocity and path efficiency computation at finer time resolution.
- **Continuous parallel sensing:** Redesigning the scan architecture to maintain constant power across all rows with simultaneous column readout, eliminating inter-row dead-time and enabling truly continuous multi-touch detection.
- **Interleaved grid pattern:** As shown by Akhtar and Kakarala [1], the interleaved electrode geometry outperforms the diamond pattern for all standard subpixel interpolation algorithms (Gaussian, Center of Mass, Linear, Parabolic) under both noiseless and noisy conditions. A future hardware revision should replace the current parallel-strip traces with an interleaved pattern to improve finger localization accuracy, particularly for the curved and diagonal finger paths characteristic of Braille reading.
- **Evaluation with visually impaired readers:** Conducting user studies with actual Braille readers who are visually impaired to obtain ground-truth behavioral data, validate metric relevance, and assess usability of the system for its intended population.

VII. Conclusion

This paper presented a low-cost, non-invasive conductive grid-based system for real-time multi-metric analysis of Braille reading behavior on a 7×7 word grid. Seven complementary metrics— rolling WPM with EMA smoothing (M-S1), velocity IQR and motor consistency (M-T2), inter-word hesitation rate (M-H1), word-level Time-on-Task (M-D2), grid skip rate (M-D3), path efficiency (V-7), and a six-panel live visualization suite (V-2 through V-7)—are computed in a multi-threaded Python pipeline with no wearable or optical components.

Experimental results on [N] participants demonstrate [summary of key accuracy and latency results]. The metrics converge on the same difficult grid cells across independent measurement axes, providing convergent validity and a rich signal for adaptive Braille tutoring. The proposed system is the first low-cost platform to simultaneously capture all of these dimensions in real time, opening new possibilities for data-driven Braille literacy instruction at scale.

Acknowledgment

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References

[1] H. Akhtar and R. Kakarala, "A methodology for evaluating accuracy of capacitive touch sensing grid patterns," *J. Display Technol.*, vol. 10, no. 8, pp. 672–682, Aug. 2014.